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ITMO UNIVERSITY

International Research and Educational Center
for Physics of Nanostructures

LAB 2
for the Course
"Adaptive and Robust Control"

on the topic:
SIMPLE EXAMPLE OF ADAPTIVE CONTROLLER DESIGN

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INTRODUCTION

0.1 Introduction

Problem statement. Consider disturbed plant

$$\dot{x} = \theta x + u + \delta \quad (2.1)$$

where δ is unknown bounded disturbance such that $|\delta(t)| \leq \bar{\delta}$. As in the work №1. x is the state variable, u is the control, θ is the unknown constant parameter.

The objective is to design a control that will provide the boundedness of all the signals in the closed-loop system and ensure the inequality

$$|x_m(t) - x(t)| \leq \Delta, \quad \forall t \geq T, \quad (2.2)$$

where Δ, T are the maximum steady-state error and transient time, respectively, $x_m(t)$ is the output of reference model (1.2). It is assumed that the constants Δ and T can be regulated by the parameters of controller.

Solution № 1. Solution obtained by applying nonlinear static feedback

$$\hat{\theta} = -\gamma x \varepsilon \quad (2.3)$$

in (1.4) instead of adaptation algorithm (1.5). Substituting (2.3) for $\hat{\theta}$ in (1.4) we obtain robust control law

$$u = \gamma x^2 \varepsilon - \lambda x + \lambda g. \quad (2.5)$$

Solution № 2. Modify adaptation algorithm (1.5) by applying the leakage factor (static linear feedback):

$$\dot{\hat{\theta}} = -\sigma \hat{\theta} - \gamma x \varepsilon, \quad (2.8)$$

where σ is the constant value. Then use this adaptive robust modification together with adjustable control (1.4).

0.2 The order of the work

The order of the work

1. Simulate the adaptive stabilizing (reference $g(t) \equiv 0$) system for the linear disturbed plant

$$\dot{x} = \theta x + u + \delta$$

consisting of the adjustable controller

$$u = -\lambda x - \hat{\theta}x,$$

adaptation algorithm

$$\dot{\hat{\theta}} = -\gamma x \varepsilon$$

$$\varepsilon = x_m - x = 0 - x = -x$$

and the disturbance

$$\delta(t) = (1+t)^{-1/8} \left[1 - \theta(1+t)^{-1/4} - \frac{3}{8}(1+t)^{-5/4} \right].$$

The controller parameters are given as $\gamma = 0.25$, $x(0) = 1$, $\hat{\theta}(0) = 1$. Separately plot variable x together with x_m , variable u and variable $\tilde{\theta} = \theta - \hat{\theta}$. Make a conclusion.

2. Repeat the previous experiment with the control (2.5) and three different gains γ . Reference signals $g(t)$ are taken from Lab1. Define the influence of the gain γ on the closed-loop system properties. Separately plot variable x together with x_m and variable u .

3. Repeat experiments 2.2 using the control (1.4) with σ -modification (2.8) and with the same three different γ . Repeat experiment for a certain γ and three different coefficients σ . Reference signals $g(t)$ are taken from Lab 1. Define the influence of the coefficients γ and σ on the closed-loop system properties. Separately plot variable x together with x_m , variable u and variable $\tilde{\theta} = \theta - \hat{\theta}$.

0.3 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

Symbol	Explanation
x	State variable
u	Control
θ	Scalar parameter
ε	Control error
x_m	Output of the reference model
g	Reference signal
λ	Parameter for transient performance
σ	Parameter for robustness

0.4 Variant

Table 2 — Objective function and constraint

Var.	θ	$f(x)$	λ	$g(t)$
15	29°	x^4	6	$\cos(2t) \sin(t) + 1$

1 SOLUTIONS

1.1 Task 1

Consider the plant:

$$\dot{x} = \theta x + u + \delta$$

where δ is an unknown bounded disturbance satisfying $|\delta(t)| \leq \bar{\delta}$. The reference model is:

$$\dot{x}_m = -\lambda x_m + \lambda g$$

The tracking error is defined as:

$$\varepsilon = x_m - x$$

The closed-loop error dynamics are:

$$\dot{\varepsilon} = -\lambda x_m + \lambda g - \theta x - u$$

The adaptive control law is designed as:

$$\begin{cases} u = -\hat{\theta}x - \lambda x + \lambda g \\ \dot{\hat{\theta}} = -\gamma x \varepsilon \end{cases}$$

Given the disturbance $\delta(t) = (1+t)^{-1/8}[1 - \theta(1+t)^{-1/4} - \frac{3}{8}(1+t)^{-5/4}]$, with $\gamma = 0.25$, $x(0) = 1$, $\hat{\theta}(0) = 1$, and $g(t) = 0$, the simulation setup is shown in Figure 1.

1.2 Simulink Model

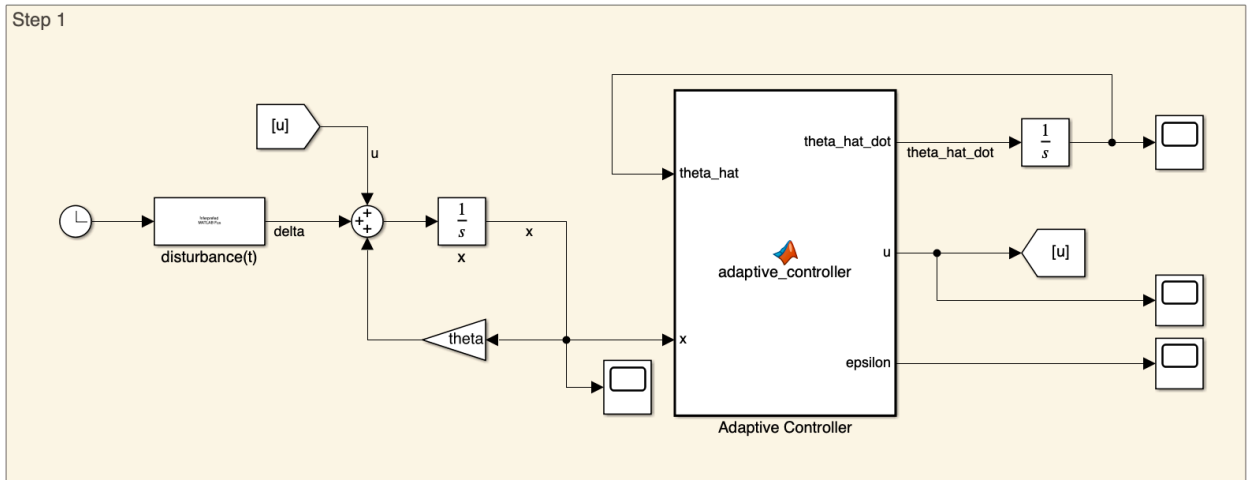
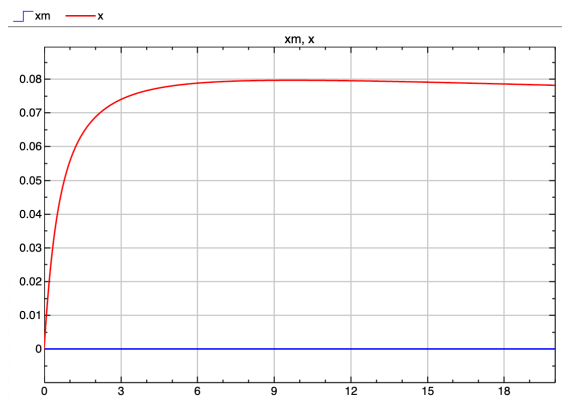
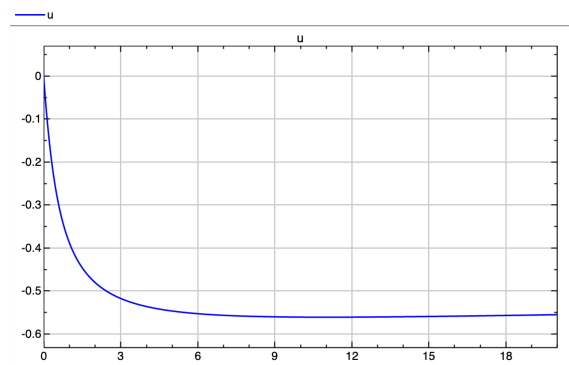


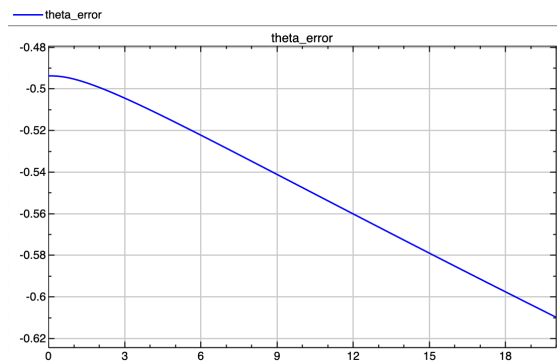
Figure 1 — Model 1



(a) Xm-X Output



(b) Control Input



(c) θ tilde Output

Figure 2 — Simulation Results for $x_m = 0$

1.2.1 Conclusion

Simulation results clearly show that when disturbance is introduced into the model, the parametric error cannot converge to zero using only the adaptation algorithm.

1.3 Task 2

The nonlinear static feedback is given by:

$$\hat{\theta} = -\gamma x \varepsilon$$

Using this feedback instead of the adaptation law gives the robust control:

$$u = \gamma x^2 \varepsilon - \lambda x + \lambda g$$

Setting $g(t) = \cos t + 4$, the plant simulation with leakage factor is shown in Figure 3.

1.4 Simulink Model

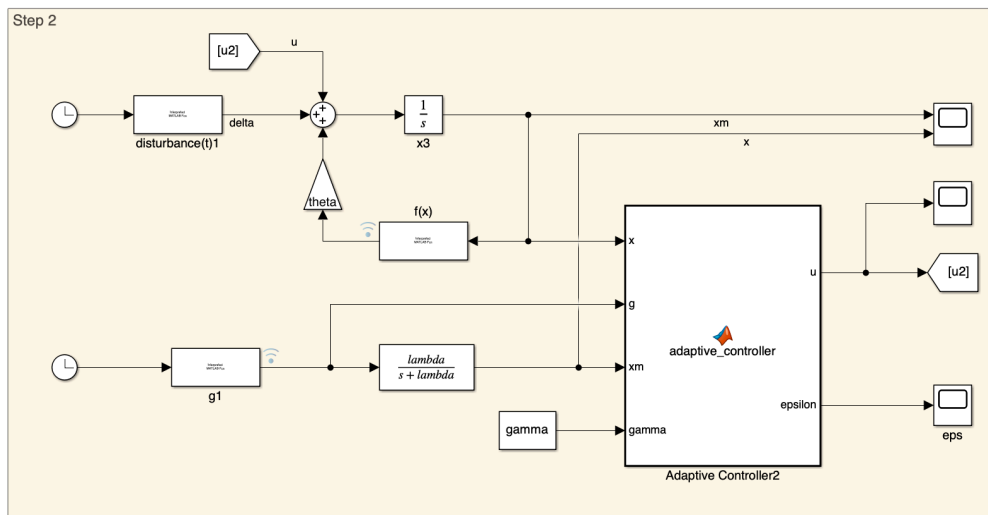
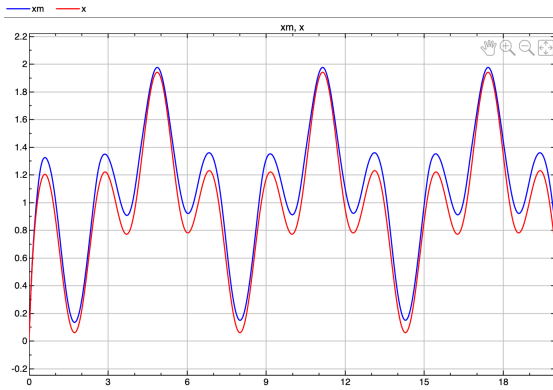
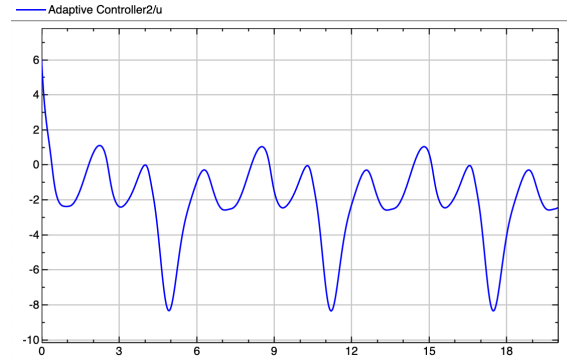


Figure 3 — Model 2

Varying the adaptation gain γ

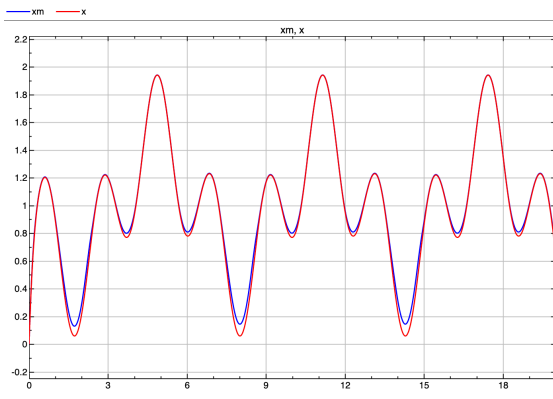


(a) Xm-X Output

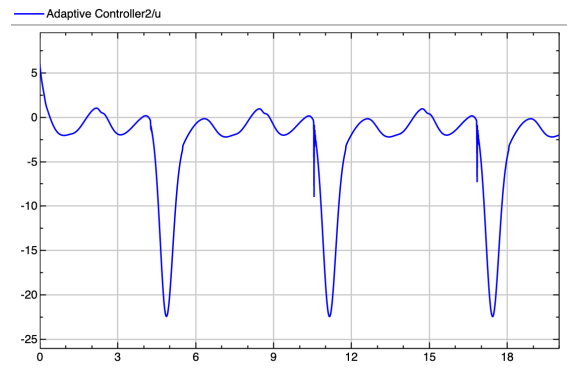


(b) Control Input

Figure 4 — $\gamma = 1$

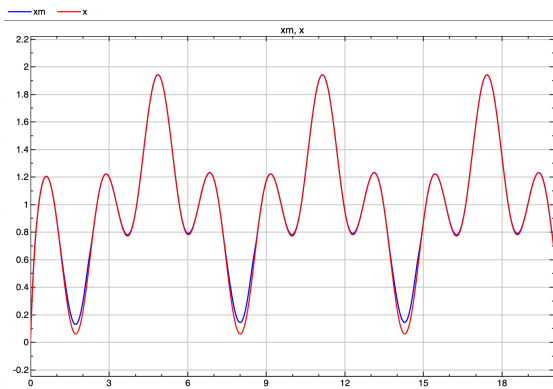


(a) Xm-X Output

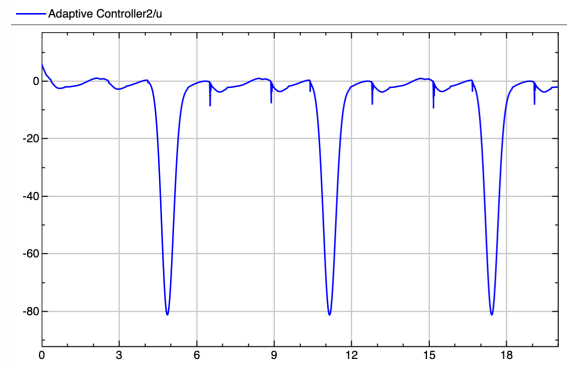


(b) Control Input

Figure 5 — $\gamma = 100$



(a) Xm-X Output



(b) Control Input

Figure 6 — $\gamma = 500$

1.4.1 Conclusion

Simulation results demonstrate that using nonlinear static feedback reduces the system parametric error compared to the adaptation algorithm alone. More-

over, a larger adaptation gain enhances system robustness and further decreases parametric error.

1.5 Task 3

The adaptation law is modified with a leakage term:

$$\dot{\hat{\theta}} = -\sigma \hat{\theta} - \gamma x \varepsilon$$

where $\sigma > 0$ is constant.

Setting $g(t) = \cos t + 4$, the plant simulation with nonlinear static feedback is shown in Figure 7.

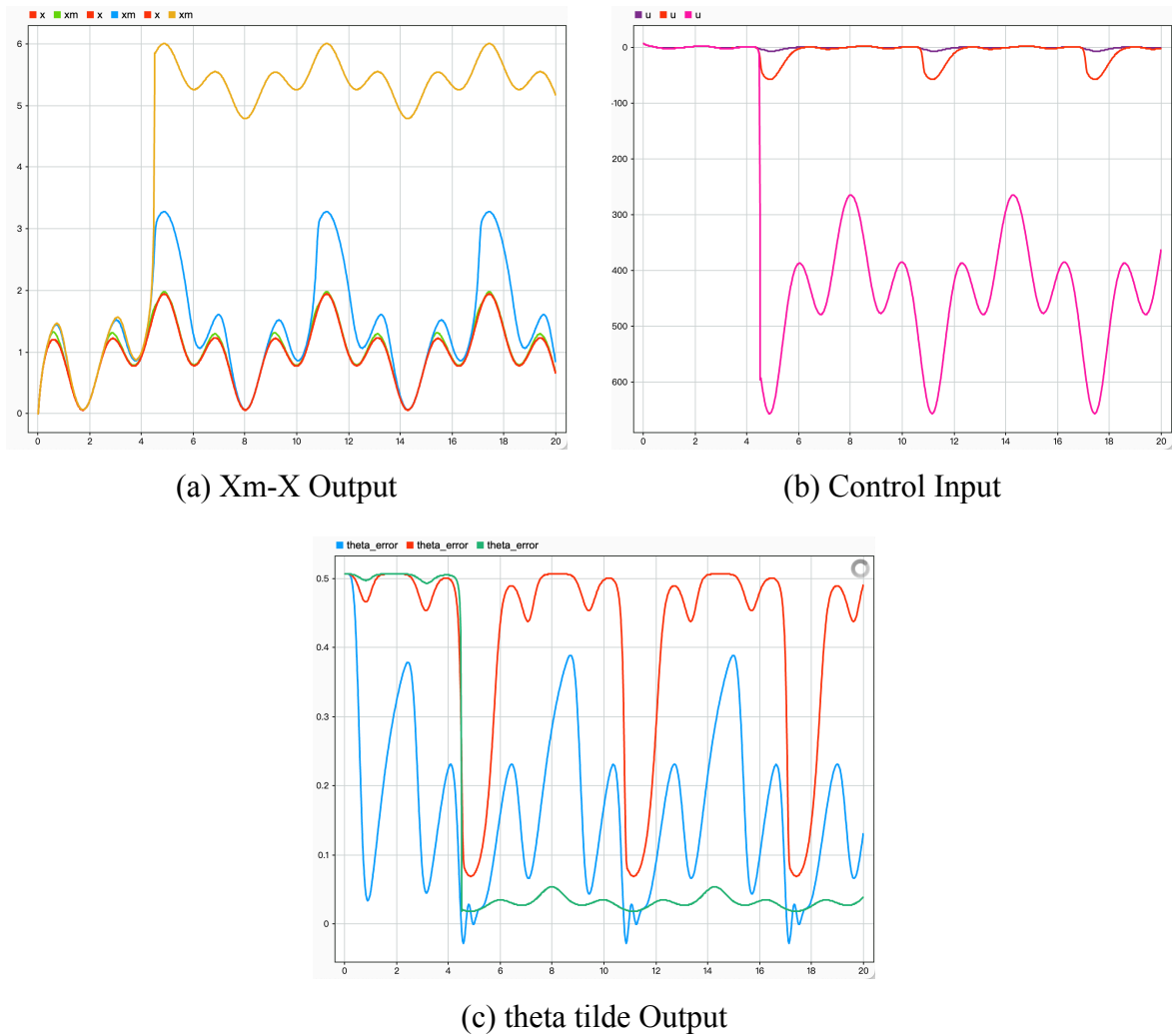


Figure 7 — Simulation Results with Leakage Factor $\sigma = 100, \sigma = 500, \sigma = 1$

1.5.1 Conclusion

Simulation results indicate that a smaller leakage factor leads to stronger system robustness and a smaller parametric error.